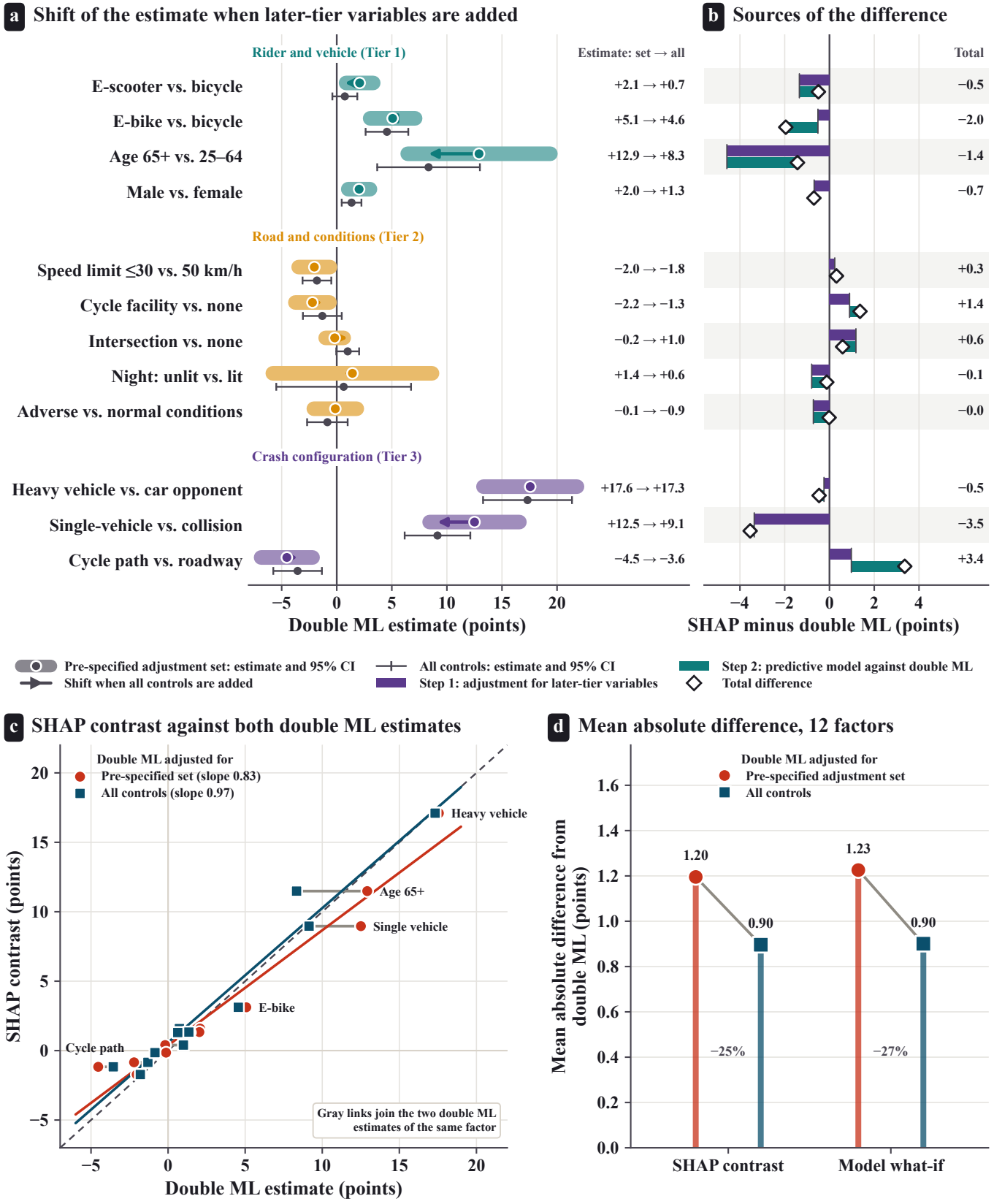
a
Shift of the estimate when later-tier variables are added
Rider and vehicle (Tier 1)
Estimate: set → all
E-scooter vs. bicycle
+2.1 → +0.7
E-bike vs. bicycle
+5.1 → +4.6
Age 65+ vs. 25–64
+12.9 → +8.3
Male vs. female
+2.0 → +1.3
Road and conditions (Tier 2)
Speed limit ≤30 vs. 50 km/h
−2.0 → −1.8
Cycle facility vs. none
−2.2 → −1.3
Intersection vs. none
−0.2 → +1.0
Night: unlit vs. lit
+1.4 → +0.6
Adverse vs. normal conditions
−0.1 → −0.9
Crash configuration (Tier 3)
Heavy vehicle vs. car opponent
+17.6 → +17.3
Single-vehicle vs. collision
+12.5 → +9.1
Cycle path vs. roadway
−4.5 → −3.6
−5 0 5 10 15 20
Double ML estimate (points)
b
Sources of the difference
Total
−0.5
−2.0
−1.4
−0.7
+0.3
+1.4
+0.6
−0.1
−0.0
−0.5
−3.5
+3.4
−4 −2 0 2 4
SHAP minus double ML (points)
Pre-specified adjustment set: estimate and 95% CI
Shift when all controls are added
All controls: estimate and 95% CI
Step 1: adjustment for later-tier variables
Step 2: predictive model against double ML
Total difference
c
SHAP contrast against both double ML estimates
Double ML adjusted for
Pre-specified set (slope 0.83)
All controls (slope 0.97)
Heavy vehicle
Age 65+
Single vehicle
E-bike
Cycle path
Gray links join the two double ML estimates of the same factor
SHAP contrast (points)
Double ML estimate (points)
d
Mean absolute difference, 12 factors
Double ML adjusted for
Pre-specified adjustment set
All controls
1.20
0.90
−25%
1.23
0.90
−27%
Mean absolute difference from double ML (points)
SHAP contrast
Model what-if